Earth-1: a deterministic, census-true civilization substrate with no language model in the loop

Pietro Novelli

Earthling Labs, London, United Kingdom

Correspondence: pietro@vaultik.com · Version 3.1 · 8 September 2026 · Epoch 3, physics tag freeze-0.9 · A pre-registered falsification protocol and its first register of misses

Abstract

Computational models of society divide into two families that trade against each other along a single axis. Classical agent-based models are cheap and inspectable but behaviourally thin: their agents follow heuristics the modeller chose. Societies of language-model agents are behaviourally rich and demographically groundable, but their cost scales with the product of agents, questions and days; their results are conditional on the particular checkpoint behind the agents; and their own authors state that outputs are best read as hypotheses to be examined empirically, “not as evidence about real societies”[1]. We describe Earth-1, a deterministic civilization model that occupies a third position. It contains no language model anywhere in its runtime path, and it can be held to a prediction it made before the outcome existed.

Agents are drawn from a census-grounded five-dimensional joint population across 194 countries — 216 cells per country, fitted by iterative proportional fitting over pooled donors, which beat an independent-marginal baseline on withheld joints by 24.8% at $p \leq 2.4 \times 10^{-13}$ — and are advanced by an eight-force behavioural grammar under a deterministic tick.

The substrate reproduces the world's material anchors, and the dynamics preserve them. At 200,000 census-true agents over 180 days the world sits at global poverty 45.9% against a real 46.1%, median income \$9.61 per day against \$9.27, crude death rate 0.0073 per year against 0.0076, and a 65-and-over adult share of 14.1% against 13.6% — with each line labelled in the paper by whether it was a calibration target, holds by construction, or is genuinely exposed. That board reproduces at twenty times the population. Determinism is exact rather than statistical: three independent execution paths converge on one identical world hash, a second internally developed implementation reproduces 38 of 38 checkpoints bitwise across a simulated year, and a perturbation of 10^{-9} applied to one agent in a four-million-agent world changes that hash and restores it exactly on undo. Against 21,776 scored item-country cells drawn from the World Values Survey (Wave 7) and the Pew Global Attitudes surveys, under leave-one-country-out, the model agrees with the real human majority on 83.5%, and 14.2% of its answers fall inside the surveys' own 95% sampling noise — reported beside the same metrics computed for the statistical baselines on the identical cell set, one of which (a regional lookup) exceeds both figures.

The architecture is what makes this affordable. State costs 981 bytes per agent, flat to 0.7% across a hundredfold range of population; the resource envelope of a hundred-million-agent run was forecast before it was executed and held on both memory and wall-clock; and one world of four million people has run continuously past day 16,860 on a single 48-core server. Serving the same world by prompting each agent once per day would have consumed roughly 40 trillion tokens.

We report the misses at the same resolution as the results, and Section 3.3 is a complete register of them. The opinion readout sits about two percentage points behind a region-copy baseline in aggregate while being outright best of four methods on 64 of 468 held-out items; the aggregate event-direction gate stands at 4 of 10 registered calls against a decision rule of 80%; and extreme poverty and unemployment both run high against their anchors. Each was measured against gates fixed by written ruling before the run that tested them. That is the property this paper is about: Earth-1's distinguishing feature is not that it is right, but that it is the kind of object that can be shown to be wrong.

Keywords: agent-based modelling; computational social science; determinism; reproducibility; pre-registration; synthetic populations; iterative proportional fitting; falsifiability

1 Main

The computational study of societies has organised itself into two research programmes that trade against each other along a single axis.

The older programme, agent-based modelling in the Schelling–Sugarscape lineage[2,3], buys tractability with behavioural poverty. Rules are hand-specified, parameters are chosen subjectively, and the interesting question — why a population does what it does — is answered by the modeller before the simulation starts. What such models gain is considerable: they are cheap, they are inspectable, and every mechanism in them can be read. The tradition has produced durable results precisely because its mechanisms are legible, from residential segregation arising out of mild individual preference[2] through threshold models of collective behaviour[4] and global cascades on random networks[5] to bounded-confidence opinion dynamics[6,7]. It has also been used operationally, most visibly in epidemic policy[8]. Its recognised weakness is not that its agents are simple but that their simplicity is a modelling choice rather than a measured fact, which makes the step from model behaviour to claims about people a matter of argument rather than evidence[9,10].

The newer programme replaces the rule with a language model. Generative agents demonstrated that language-model-driven characters could produce recognisably human routines and social coordination[11], and successive systems have scaled that design substantially. The largest reported to date instantiates one billion language-model-powered agents on a scale-free network through prompt caching, knowledge distillation to surrogate models, and mixture-of-models routing[1]. These populations are behaviourally rich in a way no hand-written rule reproduces, and they can be conditioned on real demographic distributions drawn from survey microdata.

Three properties travel with that design. Cost scales with the product of agents, questions and days, because behaviour is generated per agent per question rather than computed from state. Results are conditional on the model behind the agents: the same reported system moves its stance-change rate by up to 4.7 percentage points, with topic-dependent sign, when the communication language of the underlying model is swapped[1]. And reproducibility is statistical rather than exact — the strongest such claim in that literature is a coefficient of variation below 0.01% across five runs, which is a statement about the narrowness of a distribution, not about the recovery of a particular world.

The authors of that work are admirably direct about what follows. Their outputs, they write, “should not be treated as substitutes for human participants” and are “best read as hypotheses to be examined empirically, not as evidence about real societies”[1].

We take that disclaimer seriously, and this paper begins from the gap it opens.

1.1 The property neither family has

A model whose outputs are hypotheses cannot be falsified by the world. It can be interesting, it can be suggestive, it can generate research directions — but it cannot lose a bet, because nothing it produces was ever committed to in a form that the world could later contradict. A model that cannot be falsified accumulates capability without accumulating credence. It becomes more impressive every year without becoming more trustworthy.

Falsifiability of this kind is not a matter of scientific etiquette[12]. It is an architectural requirement with a hard precondition: to hold a model to a prediction, one must be able to recover the exact world that produced it. Not a world drawn from the same distribution — *that* world, bit for bit, months later, on different hardware. The requirement is familiar from the reproducibility literature in computational science[13,14], and it has a specific history in agent-based modelling, where independent re-implementation of published models has repeatedly failed to reproduce their results[15]. The remedies developed in response — standardised model description[16], pre-registration of analyses[17,18] — are procedural. Earth-1 takes the requirement literally and makes it architectural, and everything else this paper describes follows from that decision.

1.2 Earth-1

Earth-1 is a deterministic civilization model built on that requirement. Its population is drawn from a census-grounded five-dimensional joint distribution across 194 countries and aggregated with census weights, so that every global figure it reports is an estimate of the world rather than of the sample resident in memory. Its agents are advanced by an eight-force behavioural grammar under a deterministic tick. No language model appears anywhere in its runtime path.

Removing generative inference from the loop is not an economy measure, though the economics that follow are dramatic. It is what buys exact reproducibility, and exact reproducibility is what makes every subsequent claim in this paper checkable: a frozen forecast can be reopened and re-run, a perturbation can be contrasted against a null branch of the same world under common random numbers[19], and a result that arrives months after the run can still be attributed to the thing that caused it.

The consequence is a different relationship between the model and its own errors. A system whose behaviour is generated cannot say which of its mechanisms produced a given output; a system whose behaviour is computed from state can. Earth-1's misses resolve, one at a time, onto operators that are absent from the physics and can be named before any repair is attempted. Section 3.3 lists them. Each entry is simultaneously an admission and a testable prediction about what should change when the operator is added.

1.3 Contributions

This paper reports the following.

1. **A census-grounded population substrate.** A per-country joint over sex, age band, education, income and settlement, fitted by iterative proportional fitting[20] over pooled donors, which beats an independent-marginal baseline on withheld joints by 24.8% at $p \leq 2.4 \times 10^{-13}$. Adult age structure is derived from stable-population

density under each country's own Gompertz survival curve[21] rather than from a parametric draw (Section 2.1, Methods 4.1).

2. **Exact determinism, verified along three independent paths and one re-implementation.** Same-seed rebirth, save-and-load continuation, and flag-on baseline converge on a single world hash; a second implementation, developed internally as a laboratory rewrite, reproduces 38 of 38 checkpoints bitwise across a 365-day year; and a 10^{-9} perturbation to one agent in a four-million-agent world is detected by that hash and reversed exactly on undo (Section 2.2). A hash endpoint on the public interface makes the reproducibility claim exercisable by any key holder.
3. **A material board that matches real anchors and holds under scale.** Poverty, income, mortality and age structure land on their anchors at 200,000 agents and reproduce at four million, while a five-day census is invariant to 0.08 percentage points across a 500-fold range of population (Section 2.3, 2.4).
4. **Population-level agreement with real humans, scored against the major international surveys.** 83.5% majority agreement across 21,776 item-country cells from the World Values Survey and the Pew Global Attitudes surveys under leave-one-country-out, with 14.2% of cells inside the surveys' own sampling noise, reported alongside a four-method baseline board on which the readout is behind two of the three baselines (Section 2.5).
5. **A mechanistic account of social transmission, and a measured separation of what is designed from what emerges.** Within one generation, the heavy tail of income is shown to be drawn at genesis while wealth concentration is shown to grow (Section 2.6).
6. **A demonstrated parameter-learning loop against a hard control.** On sealed synthetic twin worlds with planted parameters, an experiential learner beats an informed non-learner given identical observations without persistence, by 4.8× on late-window score, positive in 16 of 16 well-specified worlds (Section 2.7).
7. **Retrodiction under a frozen-forecast protocol**, with the register of hits and misses reported at equal resolution (Section 2.8, 3.3).
8. **An economics that permits continuous operation.** 981 bytes per agent, flat across a hundredfold scale range; a four-million-person world running past day 16,860 on one server; roughly 40 trillion tokens of generative inference not spent (Section 2.9).

1.4 What this paper withholds

This is a description of a system and its measured behaviour, not a reproduction specification. Force update equations, fitted constant values, calibration readout formulae, injector templates, the registered vocabulary and internal module structure are proprietary and are described functionally throughout. Where a quantity would permit reconstruction of a withheld mapping, it is stated at reduced precision or omitted and the omission is marked. Every claim below rests on measured outputs against registered gates and sealed judges rather than on disclosed internals.

2 Results

2.1 Overview of Earth-1

The dominant design for large-scale social simulation instantiates *N* reasoning entities, gives each a natural-language profile and a policy realised by a language model, dispatches their interactions through an event queue, and aggregates the emitted actions into a population-level statistic. Earth-1 does not implement a smaller version of this. It does not instantiate a population of reasoners at all.

The difference is categorical rather than one of degree. There is no per-agent natural-language reasoning anywhere in Earth-1: agents do not deliberate, do not emit text, and hold no beliefs expressible as sentences. What the engine evolves is a single civilization whose population is a state manifold — a census-grounded joint distribution instantiated in memory at 981 bytes per agent, with behaviour expressed as arithmetic over that state. An individual is an addressable view of the manifold rather than a separately-reasoned process. In a swarm the agent is the unit of computation and the society is the sum of its outputs; in Earth-1 the civilization is the unit and the person is a coordinate within it.

Earth-1 carries 194 countries. Each holds a five-dimensional joint over sex (2), age band (6), education (3), income (3) and settlement (2) — 216 cells per country. Age, sex and settlement margins are census-derived for all 194. The joint itself is drawn by iterative proportional fitting over pooled donors, which beat an independent-marginal baseline by 24.8% on withheld joints at $p \leq 2.4 \times 10^{-13}$: the correlation structure between education, income and residence is measured rather than assumed away. The adult age pyramid is derived from stable-population density using each country's own Gompertz survival curve rather than a parametric draw. Every country's joint carries a declared provenance recording whether its education and income margins were measured from survey microdata or pooled from income-tier donors, so a consumer of any figure can see which countries carry it and how well grounded each one is; the provenance register appears in Section 3.3.

This is a different object from a profile pool. Where a swarm samples agent personas from a fixed set of survey records, Earth-1 instantiates a country's joint distribution and draws people from it, then attaches to every agent a weight equal to the ratio of its country's census share to its share of the simulated population. Global figures are therefore population-true against a world anchor of 8,140,897,523 people: a poverty rate reported here is an estimate of the world, not of the sample that happens to be resident in memory.

On top of the substrate sits an eight-force behavioural grammar. The forces are the channels through which circumstance becomes disposition and disposition becomes action; their definitions, couplings and constants are withheld. What matters for the results below is structural: the grammar is arithmetic over state, it is applied identically to every agent, and it is the same grammar in every country. Earth-1 does not carry per-country behavioural rules, so cross-country variation in its outputs is a consequence of cross-country variation in circumstance and composition rather than of tuning.

2.2 Determinism and exact reproducibility

The first result is a property of the instrument rather than of the world it models. Earth-1's determinism block records six checks of six passing. Three independent construction paths — rebirth from the same seed, continuation from a saved checkpoint, and a flag-on run against its own baseline — converge on one identical world hash, `49bb7d1d949f28f3`. Branch common-random-number contrast passes, historical rebirth passes, and the question-answering layer reproduces a model probability of 0.4311624 exactly on demand.

A reproducibility claim is only as strong as the sensitivity of the detector that would catch a violation, so the hash was tested in the failing direction. On the live four-million-agent world, moving a single agent's internal well-being value by 10^{-9} changed the world hash, and undoing the perturbation restored the original hash exactly. Two runs that agree on this hash therefore agree on the state of the world down to one part in a billion of one person's interior. Bit-identity here is not an artefact of a hash too blunt to disagree.

The strongest form of the claim is that the physics, not the code path, is the invariant. Against a registered tolerance of bitwise zero, a second implementation and the canonical engine produced **38 of 38 identical checkpoints across a full 365-day year at 200,000 agents**, with equal day-365 world hashes, together with 11 identical checkpoints per seed across two further seeds at a smaller population. No physics was changed to obtain any of that agreement. We state the provenance plainly: the second implementation is Earthling Labs' own laboratory rewrite of the daily path, not a third party's, so what this establishes is that the physics as specified pins the bits across two code bases under one roof. A genuinely independent port is the outstanding test, and the hash endpoint on the public interface (Methods 4.11) exists so that anyone attempting one can check the result without our involvement.

Why exact reproducibility is not the same as run-to-run stability

A coefficient of variation below 0.01% across five runs[1] establishes that a conclusion drawn from one run is not an artefact of stochastic seeding. It is worth being precise about what produces it: in the reported system, interactions at scale resolve through a surrogate precomputed into a lookup table, so the only stochasticity across runs is which influencers are sampled each round — and at millions of samples per round, a vanishing coefficient of variation is what the law of large numbers guarantees. It is a statement about the dispersion of a sampling distribution, not about the identity of an object or the fidelity of the world. Every run remains a fresh draw, so there is no artefact to freeze and consequently nothing for a pre-registration to attach to. Exact determinism supplies the missing object: a specific world, recoverable by hash, to which a claim can be pinned before its outcome is known. The distinction is structural and carries no implication about accuracy.

2.3 The material board

At 200,000 census-true agents over 180 days under the frozen physics configuration, the world's material aggregates sit on their real anchors (Table 1). We report the table with a provenance column, because its lines make three different kinds of claim and a reader is owed the distinction: some anchors were calibration targets, some hold by construction of the substrate, and some are genuinely exposed to the dynamics.

Table 1 | The material board at 200,000 census-true agents, 180 days, freeze-0.9, with the provenance of each line. *Calibrated*: the quantity, or the fitted moments that pin it, was a target or per-cycle gate of the calibration ladder. *By construction*: the quantity follows from the substrate's fitted structure. *Exposed*: the calibration does not pin it. Population-true global figures against real-world anchors.

Quantity	Earth-1	Real anchor	Provenance	Status
Poverty headcount, \$8.30/day	45.9%	46.1%	calibrated (via fitted moments)	on anchor
Median income	\$9.61/day	\$9.27/day	calibrated target	on anchor
Crude death rate	0.0073/yr	0.0076/yr	by construction	on anchor
Adult share aged 65 and over	14.1%	13.6%	by construction	on anchor
Mean age at death	69.0 yr	66.2–71.8	calibrated target	in band
Poverty headcount, \$3.00/day	16.8%	10.4%	**exposed**	1.6× over — see 3.3
Unemployment rate	∼9.7%	4.8%	level set at genesis	2× over — see 3.3

Read with the provenance column, the table makes a narrower and more defensible claim than a validation board: the substrate reproduces the anchors it was built against, and 180 days of dynamics preserve them. Preservation is not trivial — the dynamics demonstrably reshape other genesis structure, most notably the within-generation wealth distribution of Section 2.6.5 — but it is also not separately gated: every board gate is measured *at* the 180-day state, so survival-through-dynamics is inside each gated measurement rather than tested as its own stationarity law, and we say so rather than implying otherwise. The line the calibration genuinely does not pin, deep poverty, is a miss, and the unemployment level is a constructed genesis quantity that the current physics carries rather than corrects. Both failures are informative rather than diffuse: both concern the lower tail of material circumstance, and both resolve in Section 3.3 onto named absent operators rather than a general mis-fit.

2.4 Scale and scale invariance

State costs between 975.2 and 981.6 bytes per agent across the 1M, 4M, 16M, 64M and 100M rungs — flat to 0.7% across a hundredfold range (Table 2). The tick scales as $n^{1.07}$. Birth requires a consistent factor of roughly 3.2 times the final state as working memory, which is the binding constraint on a single machine rather than the steady-state footprint. At one hundred million agents, birth takes 756.7 seconds and a world-day 924.8 seconds at a peak of 324.5 GB, on one 48-core AMD EPYC 9454P.

Table 2 | Scaling ladder. Measured on one AMD EPYC 9454P (48 cores). Values marked ∼ are recorded as approximations in the source. Peak resident memory exceeds steady-state resident state because construction requires roughly 3.2× the final state as working memory.

Population	Birth	Per world-day	Bytes/agent	Peak RSS
1,000,000	—	—	975.2	—
4,000,000	—	60 s	978.7	—
16,000,000	—	—	980.4	—
64,000,000	—	—	981.6	—
100,000,000	756.7 s	924.8 s	∼980	324.5 GB

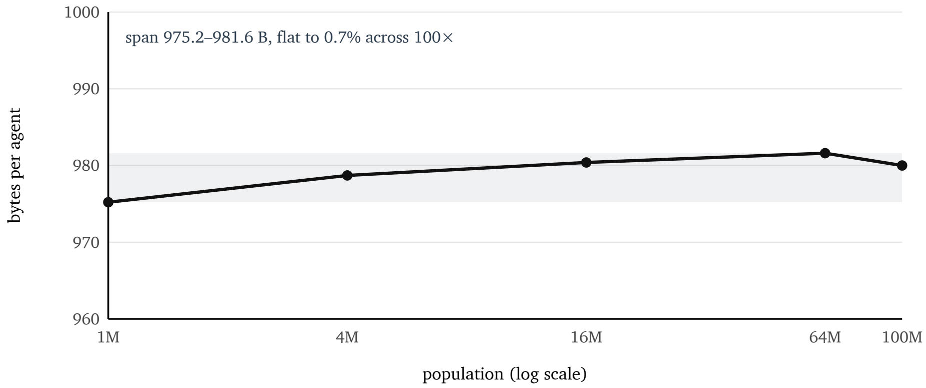

Figure 1 | State cost per agent is flat across a hundredfold range of population. Measured at five rungs on one 48-core machine. The shaded band spans the full observed range, 975.2 to 981.6 bytes per agent. Flat per-agent cost is what makes the economics of Section 2.9 hold at scale rather than degrading with population.

That ladder is a validated instrument rather than a table of observations. Extrapolating from the rungs below it, the hundred-million-agent run was forecast at **311 GB and 14.4 minutes per world-day before it was executed**, and measured 324.5 GB and 15.4 minutes. An out-of-sample engineering forecast, made beyond the largest previously measured rung, held on both axes.

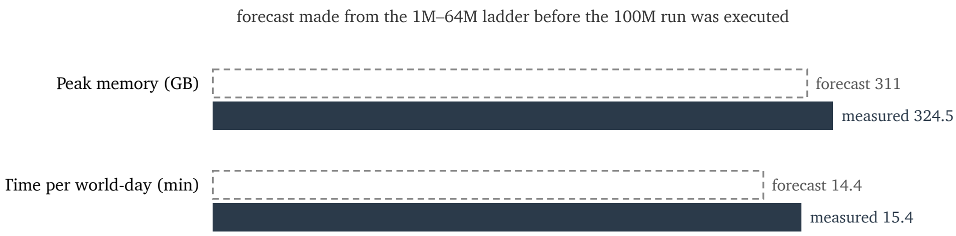

Figure 2 | The scaling laws are predictive, not merely descriptive. Peak memory and wall-clock per world-day at one hundred million agents were forecast by extrapolation from the rungs below before the run was executed, and held on both axes. An engineering forecast that survives out-of-sample use is the condition under which projections toward larger populations are admissible at all.

The board also reproduces at scale. Four million agents over 180 days on verified freeze-0.9 physics return a median income of 9.52 against the board's 9.605, a poverty headcount of 45.29% against 45.88%, a crude death rate of 0.00738 against 0.00727, and a mean age at death of 68.29 against 68.99 — every line inside its gate at twenty times the calibrated population. The extreme-poverty line moves with them, at 16.30% against the board's 16.80%, and remains high against its real anchor of 10.4%; the scale-up reproduces the board including the board's known miss. Under the frozen configuration the trajectory holds no surprise: the day-5 census is already near-calibrated — median $8.61, poverty 48.7% — and drifts gently onto the anchors over the 180 days, and through that settling window a 100M world tracks a 4M control to a poverty gap of 0.01, 0.00, 0.02 and 0.06 percentage points at days 5, 15, 30 and 60, with mean age at death identical at day 60.

A second scale result concerns invariance rather than accuracy, and its provenance needs stating precisely. The four-population control below was run on the engine's superseded default configuration rather than the frozen flags — the record initially attributed its wild census levels to equilibration and later retracted that attribution, identifying the configuration as the cause — and the levels in Table 3 are therefore comparable to nothing on any committed board. What the table establishes is the property the levels cannot touch: whatever the configuration, the census is *the same world* at every population size (Table 3).

Table 3 | Scale invariance of the day-5 census, four populations, superseded default configuration. The levels are not calibrated results and are comparable to no committed board; the result is the columns' constancy in population. Across a 500-fold range the poverty line varies by 0.08 percentage points and the median by seven cents. Under the frozen configuration the same invariance shows as the 100M-versus-4M tracking above.

Population	Median $/day	Poverty $8.30	Poverty $3.00
200,000	3.30	0.9456	0.4438
1,000,000	3.24	0.9460	0.4466
4,000,000	3.23	0.9464	0.4470
100,000,000	3.23	0.9463	0.4467

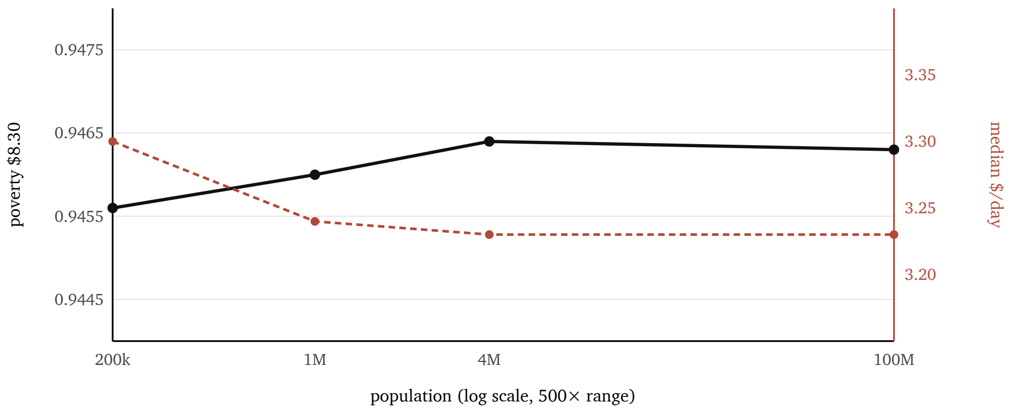

Figure 3 | The census is invariant in population size. Day-5 poverty headcount at the $8.30 line (solid, left axis) and median daily income (dashed, right axis) at four populations spanning a 500-fold range, from the superseded-configuration control of Table 3. The levels are not calibrated results; the finding is that the headcount moves by 0.08 percentage points and the median by seven cents across 500×. The calibrated board is measured at 180 days under the frozen configuration and appears in Table 1.

The board's dependence is on *time*, not on N. We are precise about how much Table 3 proves. Census weighting fixes the country mixture at every population size by construction, so aggregate invariance is partly guaranteed by the aggregation itself; what the table rules out is N-dependent drift in the within-country dynamics at the aggregate level, and what it confirms is that the population-true machinery is implemented correctly. The stronger evidence that the physics is scale-free is the previous paragraph — the full equilibrated board, dynamics included, reproducing at twenty times the calibrated population — and that is the scale result this paper leads with. One boundary travels with both: per-agent degree in the social graph is fixed, so mean-field-like aggregates are covered by these tests while network-scale phenomena such as cascades are not, and no cascade-invariance claim is made.

Scale therefore buys statistical resolution rather than a different world, and that is the affirmative case for building a larger one. Rare events — deaths, crises, tail behaviour, narrow demographic segments — become measurable in days of world-time instead of years, because a population sample can substitute for a time sample only when the larger world is demonstrably the same world. Establishing that first is what makes the exchange admissible.

2.5 Agreement with real human populations

Against 21,776 scored item-country cells across three development estates — two drawn from the World Values Survey, Wave 7 (a held-out set and a never-labelled extended set), and one from the Pew Global Attitudes surveys as compiled in GlobalOpinionQA[37] — under leave-one-country-out with named-entity abstention active and abstained cells excluded from both the model and the baselines, Earth-1 agrees with the real human majority on **83.5%** of cells. Of the 12,669 cells where the survey's own sample size is known, **14.2% are statistically indistinguishable from the survey's own 95% sampling noise** — for one cell in seven, the model's answer cannot be told apart from having asked people. Median distance is 3.78 sampling errors and mean absolute error 10.65 percentage points. The signed error is near zero overall, so the residual is family-specific tilt rather than one global bias (Table 4).

Headline metrics of this kind are only interpretable next to baselines, so we computed the same two metrics for the reconstructable baselines on the identical cell set, under Earth-1's exact decision rules (Table 4a). The naive grand-mean baseline sits below Earth-1 on both. The region-copy baseline sits *above* it on both — 84.3% agreement and 15.6% within-noise — consistent with its lower mean absolute error, and we report that plainly rather than leaving the headline pair to stand alone: on binary agreement, as on continuous error, a smooth regional lookup is currently the better estimator of a static survey margin. The per-item structure below is where the mechanistic model differs, and it is the only one of these methods whose answers can be interrogated further than the margin itself.

Table 4a | Headline agreement metrics with baselines, identical cell set and decision rules. Baseline predictions reconstructed as leave-one-country-out means of the human column (grand mean for naive; within-region for region-copy), validated against the recorded per-item error columns to within 0.01 pp on 100% of items. MrsP predictions are not retained per cell in the scoring record, so its agreement metrics cannot be computed; its pooled MAE is the cell-weighted recorded aggregate. Pooled MAE pools all three estates and is not comparable to the per-frame board of Table 5.

Method	Majority agreement	Within survey noise	Pooled MAE (pp)
Earth-1	83.5%	14.2%	10.65
Region-copy (LOO)	84.3%	15.6%	9.91
MrsP	—	—	10.74
Naive grand mean (LOO)	81.2%	12.6%	11.89

Table 4 | Distance from human majorities by question family, pooled across the two World Values Survey estates and the Pew Global Attitudes frame. Median distance is in units of the survey's own sampling error; MAE is in percentage points; signed error is positive where the model reads too high. The family classifier is itself a known limitation: a substantial block of items is family-unclassifiable and falls into 'other', and the classifier does not separate belief items from politics-of-religion items.

Family	Cells	Majority agree	Median σ	MAE (pp)	Signed (pp)
Technology and climate	735	90.5%	2.97	8.33	+0.84
Religion and values	1,163	86.0%	3.91	9.98	+0.29
Security	553	87.9%	3.65	10.24	−0.04
Democracy and governance	2,861	83.7%	3.79	10.51	+0.14
Other	14,161	83.4%	3.81	10.74	−0.03
International relations	1,162	79.7%	3.71	11.36	+0.28
Economy	1,141	79.5%	3.55	11.51	+0.21

Against continuous-valued baselines the readout is behind, and we report the board in full (Table 5). On 98 held-out World Values Survey items Earth-1 scores 11.84 pp mean absolute error against a naive grand-mean baseline's 12.80, a multilevel-regression-with-poststratification estimator's 11.18[22,23,24], and a region-copy baseline's 9.70. On 469 Pew-frame items the figures are 12.01, 12.02, 11.06 and 10.31. A replication on 110 never-labelled, never-calibrated items returns 11.20 ± 0.16 against region-copy's 9.81, so the deficit is not an artefact of item selection. Earth-1 beats the naive baseline and trails both informed baselines by roughly one to two points. The registered ACCEPT tier is 7.0 pp; this readout does not reach it.

Table 5 | Four-method national-MAE board (percentage points; lower is better). All rows are development-estate measurements. The sealed holdout and prospective judges remain unspent and are reserved for a single scored exposure under founder-witnessed final scoring, so no row here is a final benchmark result.

Frame	Items	Earth-1	MrsP	Naive	Region-copy	Recorded verdict
World Values Survey, held-out	98	11.84	11.18	12.80	9.70	miss; beats naive only
Pew Global Attitudes frame	469	12.01	11.06	12.02	10.31	miss; ≈ naive
World Values Survey, extended (never labelled)	110	11.20	—	—	9.81	replication; same verdict

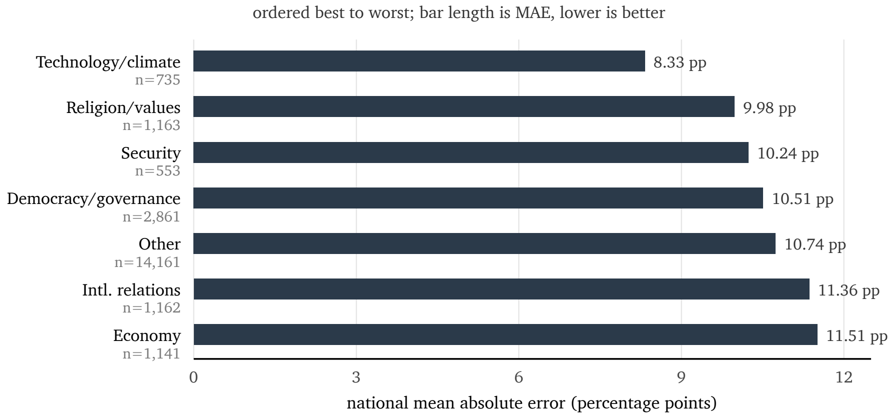

Figure 4 | National mean absolute error by question family, pooled across the World Values Survey and Pew Global Attitudes estates and ordered best to worst. Families where an attitude follows from simulated circumstance sit at the top; the two weakest families correspond to inputs the substrate does not carry, both registered in Section 3.3.

The per-item structure is where a mechanistic model earns its place. On the Pew Global Attitudes frame Earth-1 is **outright best of the four methods on 64 of 468 items and beats region-copy on 137**, and those wins concentrate in outgroup trust and material hardship — precisely the domains where an attitude follows from circumstances the engine simulates. Its losses concentrate in religion and geopolitics, which correspond to inputs the substrate does not currently carry, and both absences are registered in Section 3.3. The baselines and the model are different kinds of object — region-copy assigns a country the attitudes of its neighbours and cannot be perturbed, run forward, or queried about a counterfactual or an individual — but on the estimation task as scored, Earth-1 is behind the two informed baselines, and that is the record.

2.6 Social structure, transmission and emergence

Agent-based societies routinely claim that collective behaviour *emerges* rather than being written in, and the claim is routinely unfalsifiable. Two explanations for any neighbour-correlated pattern — genuine transmission between agents, and a shared response by similar agents to a common input — leave the same signature in aggregate statistics. A model that cannot separate them can only narrate emergence. Earth-1 can separate them, and the machinery that makes this possible is set out here.

2.6.1 A graph over people, not a channel over aggregates

Each agent holds ties across several relationship channels — household, colleagues, neighbours, friends, weak ties[25], and a small set of high-degree media hubs — with channel-specific weights and degrees. Every social computation on the daily path reads a *living view* of that graph, rebuilt after each day's deaths, migrations and job changes, so that the dead retain their edges as history while carrying no weight in current dynamics. The graph is constructed at genesis with homophilous structure: people are connected preferentially to people like them, as they are in the world being modelled. Tie weights then move continuously, and a limited amount of rewiring changes who is connected to whom.

2.6.2 Transmission is dyadic and reads a named partner's state

The daily influence operation is not a mean-field approximation. Each agent draws a small number of encounters per day, sampled by tie weight from its living neighbourhood. For each encounter, the agent's force vector moves a fixed fraction of the way toward *that specific partner's current value*, scaled by a susceptibility term depending on the agent's own condition — age, conviction, mental state, hunger. The quantity moved is read from the partner's state at that moment.

This is transmission in the strict sense: information about one person's state enters another person's state through a named edge, and the same encounters simultaneously accumulate the evidence from which conviction is updated at the end of the day. Belief and certainty co-evolve out of the same interactions. We state plainly what the operator is and is not. It moves each agent toward a partner's value and contains no term pushing an agent toward the partner's *pole*; considered in isolation it is contractive in the variance of the force distribution. Earth-1 contains no designed polarization mechanism and this paper makes no claim of emergent polarization.

2.6.3 Two forces act on every belief, every day

An agent's forces are moved each day by two competing operations. The social operation pushes the agent toward the people it encounters. A second operation computes the force state the agent's *material circumstances* imply — work, income, reserves, hunger, housing, health — and relaxes the agent toward that target. The material target is deliberately returned rather than applied, so that the social layer acts as a push and circumstance as a restoring pull.

This tension is the structural core of the model's social dynamics: an agent is moved by the people it knows and pulled back by the life it actually lives. A world with only the push saturates; a world with only the pull is a lookup table on circumstance. Because both operations are arithmetic on inspectable state, the balance between them can be read directly rather than inferred from outputs.

2.6.4 The feedback structure

Emergence requires closed loops; a one-way pipeline cannot produce it. Five are traceable on the daily path. *Influence and conviction:* encounters move forces and simultaneously accumulate agreement evidence; conviction updates from that evidence; conviction then weights susceptibility to subsequent movement. *The fabric responds to the conversation:* tie strengths update from the day's interaction, agreement strengthening a tie and disagreement weakening it, and a small number of agents find replacements closer to them in force space, so the graph the influence operator runs on tomorrow is a function of what that operator did today. *Social position feeds back into disposition:* an agent's daily deviation from its neighbourhood's mean force updates its dispositional traits, which in turn shape both susceptibility and the material force target. *Memory spreads along the graph:* events enter the world's chronicle with a scope, and that scope grows to the people who know the people it happened to, with rehearsal consolidating repeated similar events rather than stacking them. *Institutions close the loop through the population:* government policy reads aggregate population state and chooses accordingly, and the resulting policy changes the material conditions from which every agent's force target is computed.

We are careful about what the second of these supports. The *direction* of the resulting assortativity is written into the update rule, so segregation is designed; only its *magnitude*, arising from coupling with the influence dynamics, is emergent.

2.6.5 *Drawn or grown: a distinction Earth-1 can make*

The discipline that matters is refusing to credit emergence for anything that was drawn at genesis. Earth-1 is unusual in being able to tell the difference, and the demonstration is worth stating precisely.

The population's income distribution is heavy-tailed, and it would be easy to attribute that tail to correlated shocks travelling through the firm structure and the social graph. Measurement says otherwise: wage kurtosis is 247.01 at birth and 246.76 after 120 days of living. Living changes it by roughly a quarter of one percent. **The heavy tail of income is drawn, not grown, and Earth-1 says so.**

Over the same 120 days, the ratio of the 99th percentile of accumulated wealth to the median rises from 11.88 to 29.48, and the share held by the top one percent rises from 0.122 to 0.194 — with only 6.5% of agents having experienced any job loss at all. Nothing placed that concentration in the world at genesis. **Within a generation, wealth concentration is emergent, and Earth-1 says that too.**

An assertion of that kind requires its null, and the null has been run. The obvious deflationary account is that any accumulation process on a heavy-tailed income draw concentrates in its first 120 days, shocks or no shocks. So the full shock-free null family was computed analytically on the untouched genesis arrays: every agent keeps its job and accumulates at a constant rate proportional to its wage, across the entire range of accumulation rates from zero to the infinite-rate limit in which the wealth distribution becomes the wage distribution itself. That family spans a p99-to-median ratio of 11.90 to 17.63 and a top-one-percent share of 0.116 to 0.141 (Extended Data Table 12). The measured values sit above *every member*: 29.48 is 1.67 times the family's ceiling, and the structural reason is decisive — under constant wage-proportional accumulation the wealth distribution converges toward the wage distribution's shape, whose own ratio is 17.6, so *no constant accumulation rate can reach the measured concentration*. The strongest shock-free competitor, a supplementary variant in which saving is proportional to income above subsistence, reaches 26.15 and 0.162 in its limit; the measured values exceed it by 13% and 20%. The concentration is not the income draw echoing through a savings identity. It requires the dynamics — correlated employment shocks, scarring, and the differential capacity of households to absorb them. One caveat travels with this result: the genesis behind the published figures was identified by reproducing the recorded day-0 concentration to 0.13% rather than from a documentary record of the run's seed, and the identification is stated as such.

Two superficially similar inequalities in the same population, one designed and one grown, correctly distinguished. This separation is unavailable to a model whose runs differ from each other, because it requires holding an entire world fixed and asking what a single mechanism contributed. We also state its boundary: across generations no mechanism in the current engine creates new mass in the income tail, since wage is carried forward at replacement without mobility and accumulated wealth restarts. Long-run inequality in this model is not grown, and is not presented as such.

2.6.6 *Emergence as a measurement*

Determinism is what makes the preceding subsection possible. Because the world is bit-reproducible, a perturbed branch runs against a null branch of the *same* world under common random numbers[19], and any difference between them is attributable to the perturbation rather than to sampling. An emergent claim in Earth-1 therefore takes the form of a paired difference with a seed-level error bar rather than a description of what the model appeared to do.

The instrument also returns negative results, which is the harder test. When a fiscal shock was applied to a single country at full fidelity, the shocked branch fell below the registered class noise floor relative to its control, and the register names the operator whose absence explains it — a null with an address, where a system that narrates emergence would have produced a story about resilience. The decomposition of the remaining neighbour correlation into transmission and homophily is quantified by a zero-transmission ablation on the schedule, using exactly the branch-versus-null apparatus described here.

The general principle is the one this section has been building toward. Earth-1 does not ask to be believed about emergence. It exposes the mechanism as arithmetic, holds the world still, and measures what each part of it contributed.

2.7 Learning from experience under sealed controls

Earth-1's parameters can be recovered from observation, and the gates for that claim were built to catch the loop pretending to learn. We state the scope precisely at the outset: this is a **synthetic-recovery experiment**, not a claim about learning from the real world. Experience Loop v0.2 ran twenty simulated twin worlds at 20,000 agents against sealed planted parameters — sixteen well-specified and four deliberately misspecified — over twenty-four cycles of thirty simulated days, with the design pre-registered before any cycle ran.

The decisive arm was added as a binding condition on the pre-registration: an *informed non-learner* with identical particle machinery and identical observation access to the experiential learner, but no persistent update. Per cycle its weights are computed fresh from that window's observation distance alone, with no multiplicative accumulation, no resampling, and parameters fixed at their prior draws forever. It may condition on what it has just seen; it may not learn. Any advantage the experiential arm shows over it must therefore come from persistence rather than from information.

Table 6 | Experience Loop v0.2 pre-registered control arms. Paired log-CRPS[26] differences stated in the direction the record states them, so positive favours the experiential learner. All arms run on the same twenty sealed synthetic worlds. The success condition was the conjunction of every gate, registered before the run.

Control arm	What it isolates	Δ log-CRPS	95% CI	p
Informed non-learner (primary)	persistence vs state conditioning	+1.51	[1.21, 1.81]	3×10^{-5}
Frozen model	any adaptation at all	+2.16	—	$< 10^{-4}$
Naive updater	structure vs smoothing	+0.55	[0.32, 0.77]	2×10^{-4}
Shuffled-resolution placebo	real signal vs spurious fit	+2.08	—	3×10^{-5}
Shock windows	adaptation when it matters	+0.59	—	10^{-4}

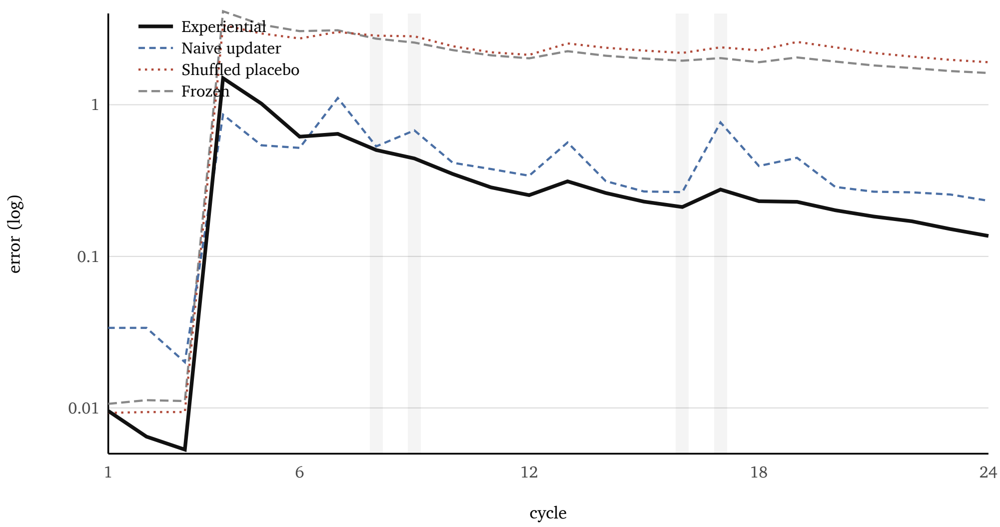

Figure 5 | Learning against its controls across twenty-four cycles on sixteen well-specified sealed synthetic worlds; lower is better, log scale. The experiential learner separates from every control and stays separated. The shuffled-resolution placebo tracks the frozen model, which is the control that matters: when the correspondence between observation and outcome is destroyed, no apparent learning appears. Shaded columns are the registered shock windows.

The hardest fair comparison is the naive updater — a structure-free smoother given the same observations — and the learner beats it by +0.55 log-CRPS [0.32, 0.77] at $p = 2 \times 10^{-4}$, roughly a 1.7-fold gap on final-cycle skill. The non-learner arm then isolates *what kind* of improvement that is: against identical machinery with persistence removed, the gap widens to a late-window score of 0.216 against 1.04 — 4.8-fold, positive in **16 of 16 well-specified worlds** — so the advantage comes from accumulated parameter learning rather than from conditioning on the current window. The causal control holds in both directions, which is the part that matters: the placebo learner is statistically indistinguishable from the frozen model ($\Delta = 0.078$, $p = 0.90$), so apparent improvement does not appear when the signal it would have to come from has been destroyed. Learning appears when, and only when, there is something real to learn from. Ninety-percent intervals cover at 0.963, the loop stays honest under deliberate misspecification in four cases of four, and replay from the hash-chained ledger is bit-identical.

One apparatus note is carried in full. The scoring stage as first run omitted the non-learner arm, and the two gates involving it were computed from the arm files by the same paired method as a workaround. The registered scorer was extended to include that arm — committed the same day, fifty minutes after the stored report was emitted, so the report predates the fix. Everything recomputable from the stored report reproduces the published gates exactly (fifteen of fifteen consistency checks, including the frozen, naive, shock-window and placebo gates to the reported precision), and the stored learner curve reproduces the published late-window mean to three decimals, making the non-learner ratio arithmetically consistent with the published interval. A full re-emission of the report under the extended scorer requires the archived arm files on the host that holds them, and is scheduled.

2.8 Retrodiction under a frozen forecast

Every historical result below was produced by birthing a world at date T on real archive material, under a mechanical assertion that no input, anchor or news item post-dates T, then freezing and committing the output *before* the judge was fetched. The judge is registered beforehand by name and location with its values unfetched, and the result is scored once.

On a pre-committed register of protest-onset countries against quiet controls, Earth-1's unforced country states rank the real protest-prone countries above the quiet ones at Spearman $\rho = 0.552$ ($p = 0.0052$), with a thirteenfold discrimination margin — mean onset probability 0.417 for the positives against 0.031 for the controls, and controls at approximately zero satisfying the registered placebo criterion. That was the register's recorded result. Re-scored for this version with a wealth covariate, it downgrades, and we publish the downgrade: the register's classes are nearly collinear with World Bank income group ($\rho = -0.815$ between outcome and income), income group alone orders the model's onset probabilities at $\rho = -0.464$, and the partial correlation controlling for income falls to 0.339 with an exact within-stratum permutation p of 0.18 — not significant, on a design where positives and controls share only one income stratum and the minimum attainable p is 0.011 regardless of effect size (Extended Data Table 13). Two facts cut the other way and are reported with the same weight: within the twelve positives, income does not predict the model's score at all ($\rho = 0.056$), and Chile outranks every rich control — so the engine is not simply rank-ordering poverty. The honest verdict is that the register as constituted cannot demonstrate discrimination beyond wealth in either direction, the pooled $\rho = 0.552$ is downgraded to *confounded with income group, unresolved at this n*, and the registered fix is an income-matched control set of quiet lower- and middle-income countries. The probabilities were already known not to be calibrated as forecasts, the Brier[27] improvement over base rate being non-significant.

Table 7 | Frozen-forecast retrodiction battery. Each world was born at T on archive material only, its output frozen and committed, and the judge fetched afterwards and scored once. Model probabilities are reported to four decimal places; the branch distances that generate them are withheld.

Event	T	Registered judge	Result
Arab Spring	2010-12-16	Protest events by country, 90 days, 185-country overlap	Material-stress geography $\rho = +0.168$, $p = 0.022$; fear channel +0.069, not significant
Global financial crisis	2008-09-14	Real 2008→2009 unemployment delta	Response 0.64× real, within registered 2× band; direction right, $t = 2.2$ at 16 seeds
COVID-19	2020-02-28	Real unemployment delta	Response 3.5× real, within registered 5× band; direction $t \approx 18$
SVB banking stress	2023-03-08	Contagion to ≥ 2 further banks within 30 days	0.3019; resolved YES. Under-called; every material line abstained
Sri Lanka	2022-03-31	Government fall within 120 days	0.3985; government fell. Under-called
Truss gilt crisis	2022-09-22	UK fiscal-stress response at full fidelity	Physics abstention: shocked branch below registered class noise floor relative to control

The aggregate event-direction gate stands at **4 of 10 registered observable-level direction calls** — two of four on COVID, two of three on the financial crisis, none of three on the Arab Spring — scored on a five-replicate battery covering three events. Against the freeze decision rule of 80% this is recorded as a failure, and the freeze carries it. As evidence about the model, however, the battery is close to uninformative: the exact 95% binomial interval on 4 of 10 runs from 12% to 74%, which cannot distinguish the engine from a coin flip or from a strong forecaster, and a later sixteen-seed re-run of the financial-crisis event alone moved that call's significance from $t = 0.7$ to $t = 2.2$ — direct evidence that the five-replicate battery was underpowered. The registered closure path is a re-run of the full battery at the current seed count; until it runs, the honest statement is that the direction gate failed its decision rule on a battery too small to measure the thing it gates.

Two of the individual misses are under-calls rather than misdirections. The model put 0.3019 on banking contagion and 0.3985 on a government falling, and both happened. In the banking case every material line abstained while the United States emerged as the top force mover at twice the runner-up — right about the fear, missing the plumbing.

2.9 The economics of a language-model-free civilization

The runtime path contains zero language-model calls. We are careful about what that contrasts with, because the comparison class also runs without a language model in the loop at scale: in the largest reported system, a frontier model resolves a few hundred thousand interactions per topic, a surrogate is distilled from them, and the surrogate is precomputed into a lookup table that serves the billion-agent run[1]. The economically meaningful distinction is therefore not the presence of inference at runtime but the *provenance of behaviour*: distilled from a checkpoint, or computed from grounded state. Distilled behaviour inherits the checkpoint's conditionality — the comparison system's own measurements show absolute trust levels shifting with the model behind the agents and a stance-change rate moving 4.7 percentage points with the communication language[1] — and no output of it can be attributed to a mechanism, because the mechanism is the checkpoint. Behaviour computed from state has neither property, and it also removes the distillation pipeline itself: nothing to sample, train, validate against a teacher, or re-derive when the teacher is retired. The direct costs are measurable (Table 8); the per-agent-prompt column is the naive regime, included as the outer bound of what the loop avoids.

Table 8 | Cost of the living world. The right-hand column is the naive per-agent-prompt regime — one small prompt per agent per day at 500 input and 100 output tokens, priced at $1 per million tokens, below frontier pricing — included as the outer bound of what the architecture avoids; distilled-surrogate regimes sit between the columns, at the cost of the conditionality discussed in the text. Earth-1's own figures are measured on one 48-core machine.

Quantity	Earth-1, measured	One prompt per agent per day
Tokens per world-day (4M agents)	0	2,400,000,000
Cost per world-day	—	≈ $2,400
Elapsed world-days	16,860	16,860
To reproduce the world's full history	11.7 days on one box	40.3 trillion tokens, ≈ $40,000,000
Resident state, 4M agents	3.9 GB	—

The figure that matters is not the per-day cost but the last row of the comparison. Reproducibility that costs forty million dollars to exercise is not reproducibility anyone will exercise. An entire civilization's history that can be regenerated from scratch in under twelve days on a single machine makes bit-exact replay of a decades-long world an operational reality rather than a theoretical property, and it is what allows a claim made about day 400 to be rechecked at day 16,000 without renegotiating a compute budget.

A second consequence is temporal stability. Because no component of the pipeline — runtime or upstream — depends on a third-party checkpoint, results cannot drift when a vendor ships a new model or retires an old one, and there is no teacher whose retirement would orphan the distilled behaviour. An Earth-1 result obtained today reproduces bit-for-bit next year, which is the precondition for the frozen-forecast protocol of Section 2.8 to mean anything at all.

Scale invariance and census weighting together settle what a larger population is actually for. Because the census is invariant across a five-hundredfold range, aggregate accuracy is already population-true at two hundred thousand agents; what more agents buy is individual and subnational resolution, and the statistical power to measure rare events in days of world-time rather than years. The economic argument is finally a simple one: removing generative inference from the runtime path is what makes a civilization cheap enough to leave running, cheap enough to re-derive from scratch, and stable enough that a result obtained today can be reproduced exactly a year from now on different hardware.

3 Discussion

3.1 A model of its own

The results above describe a system that is not a cheaper approximation of a language-model society and not a larger instance of a classical agent-based model. It is a third object, and the case for treating it as its own class rests on three properties that do not co-occur elsewhere.

The first is that its population is an estimate of a real one. A per-country joint over five demographic dimensions, fitted so that the correlation structure between education, income and residence is measured rather than assumed, and aggregated by census weight against a world population anchor, means that a global figure Earth-1 reports is a claim about humanity rather than about a convenience sample. Classical agent-based models typically do not carry a demographic frame at all; language-model societies increasingly do, but they sample personas from a pool rather than instantiating a joint distribution and drawing from it.

The second is that behaviour is computed rather than generated, which is what makes the model's mechanisms addressable. When Earth-1 is wrong, the error can be traced to an operator, and when an operator is absent the register names it — the miss has an address before any repair is attempted. This is the difference between a system whose errors are enumerable and one whose errors are diffuse. A model that is wrong without being wrong about anything in particular offers no handles for repair; Earth-1's register in Section 3.3 is a list of handles, each of which constitutes a prediction about what should change when the corresponding channel is added.

The third is exact reproducibility, which converts the first two from design intentions into measurable properties. Without it, a claim that a given inequality is emergent rather than drawn cannot be tested, because the comparison requires holding an entire world fixed while a single mechanism is varied. With it, that comparison is routine.

Together these support a specific and limited sense in which Earth-1 is a foundational model — not a pretrained network, but a single load-bearing substrate and grammar onto which many domain-specific populations can be conditioned. The same world serves a question about attitudes, a question about material hardship, and a question about the geography of unrest, without being rebuilt for each; and because the substrate is shared, the marginal cost of an additional question is the cost of reading the world rather than the cost of re-simulating it. Under a per-agent-prompt regime, cost scales with questions multiplied by agents multiplied by days; under a shared-substrate regime it does not.

We are precise about where Earth-1 is not ahead. Its largest measured configuration is one hundred million agents, an order of magnitude below the billion-agent systems in the comparison class[1]. Scale is not the axis on which this work competes, and the case for a larger Earth-1 rests on resolution and rare-event power rather than on aggregate accuracy, which Section 2.4 shows is already population-true at two hundred thousand agents.

3.2 Falsifiability as an architectural property

A simulation output is not automatically evidence. For it to bear on a claim about the world, it must have been possible for the run to be wrong in a manner specified before the answer was known. Most properties commonly reported for large social simulations — population size, throughput, behavioural richness, run-to-run stability — are orthogonal to that requirement. Earth-1 treats it as the primary design constraint, from which the engineering choices follow rather than the reverse.

The argument runs in four links, each inert without the one before it. *Determinism*: identical inputs yield an identical world, verified along three independent execution paths and one independent re-implementation. *A frozen artefact*: because a run reproduces exactly, its output can be committed as a fixed object with a hash, so that "the model said *X*" becomes a checkable assertion about a particular artefact rather than a summary of a distribution that would have to be regenerated, and might regenerate differently, in order to be inspected. *Pre-registration*: once the artefact is fixed, the claim, the gates and the scoring procedure can be written and sealed while the answer is still unknown[17,18]. *A judge fetched afterwards*: only at that point is outcome data retrieved, from a source registered beforehand by name and location.

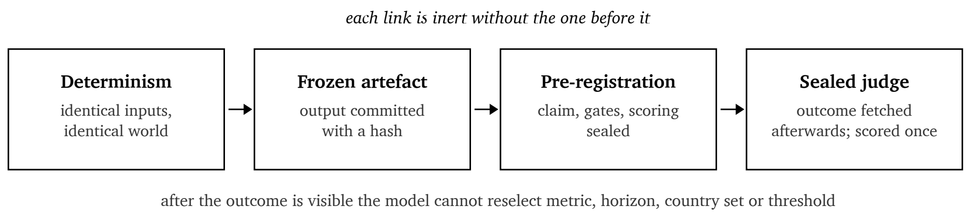

Figure 6 | The chain from determinism to evidence. Exact reproducibility yields an artefact that can be frozen; a frozen artefact is something a pre-registration can attach to; a pre-registered claim can be scored by a judge whose values were unknown when the claim was made. Statistical run-to-run stability supplies no artefact and therefore cannot enter this chain.

It is the ordering, not any individual step, that does the work. Under this sequence the model has no move available after the outcome is visible: it cannot reselect the metric, the horizon, the country set or the threshold. Retrospective scoring leaves the space of admissible framings open until after the answer is in hand, and a sufficiently large space always contains a favourable framing[28].

We also state the chain's current boundary, because it is the largest legitimate objection to this paper: as of this version, every link is verifiable end-to-end only inside Earthling Labs' own ledger. The physics is withheld, the second implementation is internal, registrations carry no external timestamp, and final scoring of the sealed estates is witnessed by the founder. Falsifiability that only the falsifiable party can observe is a protocol, not yet a public property. The programme that converts it is concrete and mostly packaging of artefacts already held: public timestamped commitments of each frozen-forecast hash and configuration stamp; release of the per-cell predictions, judge values and scoring code so every number in Sections 2.5 and 2.8 is recomputable; the hash endpoint now on the public interface, so the determinism claim is exercisable by any key holder; an evaluator-held test set scored by a named external group; prospective forecasts on a platform with its own resolution rules; and a named external witness replacing founder witnessing when the sealed estates are spent. The first three exist or are staged at this version; the other three require counterparties and are named here as commitments rather than reported as done.

A chain of this kind fails through procedure rather than theory, so the procedure is codified: one named change per cycle, so that any movement in the gates is attributable; gates fixed by written ruling before the run; VOID reserved for defects in the measuring instrument and never invoked for an unwelcome result; sealed judges registered with values unfetched until the model output is committed. Such rules are only tested when they cost something. The physics was frozen as a *freeze-with-named-regression* — carrying the failed direction gate and five named regressions on the tag rather than a fix that would have reopened the loop — and a registered substrate defect was likewise left deliberately unpatched and scheduled as a later cycle, because the physics was frozen. Both decisions are locally uncomfortable and structurally necessary. A gate that may be revised once the result is visible is not a gate.

3.2.1 A worked example: the 2022 grain-shock counterfactual

The counterfactual was frozen and committed before any outcome data was fetched, against a judge registered in advance: an acute-food-insecurity database restricted to the 48 countries assessed in both 2021 and 2022. The model lost on all three registered dimensions. Magnitude overshot by 8.5× (+326M modelled against +38.5M real). Geography carried no signal, at Spearman −0.137. Top-five overlap was one of five.

Sealed before the fetch, the written prior had predicted all three failures and named the mechanism responsible: no substitution, no subsidies, no stock drawdown. Reality then confirmed the named damper directly — Egypt's real change was exactly zero, absorbed by its bread subsidy, which is the classical entitlement mechanism by which food availability and food access come apart[29]. The run is therefore, simultaneously and without contradiction, a triple failure against its own registered gates and a successful test of a stated mechanistic hypothesis. Only the ordering makes the second reading admissible. Had the prior been written after the numbers were known, it would be narrative rather than prediction, and the identical text would carry no evidential weight whatever.

3.3 Limitations: the complete register

This section is the complete register of defects standing against Earth-1 at freeze-0.9. Each entry names one thing: a channel the physics does not contain, a fault the engine's own checks found in its apparatus, a boundary of the substrate, or a score below a gate fixed by written ruling before the run. Each is stated once, with its measured consequence and the closure path the record names. The register is published rather than summarised because a model whose errors can be enumerated this precisely is demonstrating instrument quality, and because each entry is a testable prediction about what should change when the corresponding operator is added.

Table 9 | Named open defects at freeze-0.9, with measured consequence and closure path. Absent channels are physics that has not been built; substrate boundaries are properties of the population frame; instrument defects are faults the engine's own checks found in its measuring apparatus.

Class	Defect	Measured consequence	Closure
Absent channel	No unemployment-insurance operator	Unemployment ~9.7% against real 4.8%; COVID unemployment response 3.5× the real delta	Insurance and furlough operators
Absent channel	No substitution, subsidy or stock-drawdown operators	Food shock overshoots 8.5×; geography $\rho = -0.137$; top-5 overlap 1/5	Price substitution, state transfer, inventory buffering
Absent channel	No interbank or deposit channel	Banking contagion under-called at 0.3019 on an outcome that resolved YES	Interbank exposure and deposit-flight channel
Absent channel	No bond or pension channel	UK gilt crisis invisible; branch below class noise floor relative to control	Sovereign-debt and pension-liability channel
Absent input	No national religiosity input	Religion and values is the worst opinion family at 3.91 sampling errors	Country-level religiosity series
Absent input	No geopolitical-alignment input	International relations is the worst Pew family: 11.36 pp MAE, 79.7% majority agreement	Alignment and bloc-membership series
Instrument defect	Settlement flag inverted in the current substrate	Model settlement reads as exactly one minus census in every country checked; two physics consumers affected	Registered, deliberately unpatched under the freeze; scheduled v1.1
Instrument defect	Chronicle cost	A world warmed on ninety days of real news ticks far slower than a cold one; memory spread dominates the tick	Consolidation pass; registered pre-scale cycle
Instrument defect	Encounter stream keyed on simulation day	Seed replicates do not resample the encounter sequence, so error bars on opinion observables under-sample the influence channel	Reseed the stream; quantify the under-sampling
Substrate boundary	131 of 194 country joints are tier-fallback	Education and income margins pooled from income-tier donors rather than measured from microdata	Additional survey microdata licences
Substrate boundary	No subnational geography; no household frame	No adm1 resolution; households not represented as units	National statistical tables or a negotiated source
Score below gate	Event-direction gate	4 of 10 registered direction calls; pre-registered tiers 75% ACCEPT / 85% GOOD	Election and shock injector v2
Score below gate	Opinion readout against informed baselines	11.84 pp against MrsP 11.18 and region-copy 9.70; registered ACCEPT tier is 7.0 pp	Missing inputs above; readout revision
Score below gate	Extreme poverty	16.8% at \$3.00/day against a real 10.4%, 1.6× over; persists at 20× scale	Lower-tail income and transfer operators

Four further boundaries belong on the record because they bound how the results above may be read, and none of them is a defect in the engine.

The sealed judges are unspent. Every opinion and benchmark figure in this paper is a development-estate measurement. The holdout and prospective estates — a held-out Pew Global Attitudes survey wave, a sealed judge split of the Pew-derived compilation, latest-wave survey estates, and prospective event resolutions — are registered, sealed, and reserved for a single scored exposure under witnessed final scoring. No number reported here is a final benchmark result, and the paper should not be read as one.

The learning result is synthetic. Section 2.7 demonstrates parameter recovery on twenty simulated twin worlds against sealed planted parameters. It establishes that the loop learns rather than conditions, and that it does not learn when the signal is destroyed. It does not establish learning from real-world observation, and the same freeze package records the external real-data scoreboard as a miss.

Some strong results are ordering results, not calibration results — and one has been downgraded by our own re-scoring. The protest register's $\rho = 0.552$ is confounded with income group and unresolved at its current size (Section 2.8, Extended Data Table 13); the registered fix is an income-matched control set. Its Brier improvement over base rate was already non-significant. Earth-1 is currently better at saying *where* than at saying *how likely*, and on this register it has not yet shown that *where* means more than *poor*.

Substrate provenance does not change the model's relative position. Split by provenance class, Earth-1's gap to the region-copy baseline is +0.77 pp on the 63 survey-measured countries and +0.49 pp on the tier-fallback countries present in the cycle, with the ordering preserved under estate matching; the tier-fallback countries are, if anything, slightly easier for the engine relative to both baselines. The 131-country fallback boundary is therefore a grounding limitation, not a hidden accuracy cliff — though only 22 of the 131 appear in the scored cycle, and tier-fallback cells abstain at more than twice the survey-measured rate.

Opinion performance varies sharply by region. The best region sits at 2.28 sampling errors and the worst, East Asia, at 4.66, with the farthest individual countries clustered in East and Southeast Asia. Regional structure of this size is not captured by the family-level breakdown in Table 4, and no measured input currently explains it. Two candidates are registered as hypotheses rather than findings — media environment and political-system type, neither of which the substrate carries — and if neither closes it, the closure path for this entry is genuinely unknown, which the register records as such rather than papering over.

No counterfactual has yet passed its registered gates. The grain-shock counterfactual failed all three of its dimensions and the fiscal-shock case returned an abstention, so the capability of Section 3.2.1 is, on the current record, demonstrated as protocol rather than as accuracy: the machinery for losing a bet cleanly exists and has been exercised; the machinery has not yet produced a won bet. The first counterfactual to pass under sealed protocol is the single most valuable result this programme can produce, and the mechanism cycles opened by the losses — substitution, subsidies, stock drawdown — are the registered route to it.

Finally, one methodological caveat about how the record itself is maintained. Where two committed artefacts disagree — as two files do on the number of cells abstained in the distance run — the disagreement is carried forward rather than silently reconciled, on the principle that a register which quietly harmonises its own sources has stopped being a register.

3.4 Outlook

The defects in Table 9 are not a backlog so much as a research programme with the order of work already fixed. Each absent channel names a mechanism whose addition makes a specific, quantified prediction: adding an unemployment-insurance operator should move the unemployment line toward 4.8% and shrink the COVID overshoot below 3.5×; adding substitution and subsidy operators should reduce the food-shock magnitude error from 8.5× and, if the mechanism named in the sealed prior is the right one, should recover the countries whose real change was absorbed by domestic transfers. Because the gates were fixed before the runs that failed them, those predictions can be scored the same way.

Four lines of work follow directly from results reported here. The zero-transmission ablation described in Section 2.6.6 will decompose neighbour correlation into transmission and homophily, converting the paper's structural claim about dyadic influence into a quantity — the same discipline that the accumulation null of Section 2.6.5 has now applied to the wealth-concentration claim, and that the income re-scoring of Section 2.8 has now applied, at the claim's expense, to the protest register. The protest register itself will be re-registered with income-matched controls, since its current construction cannot answer its own question. Reseeding the encounter stream will establish how much the current error bars on opinion observables under-state variance in the influence channel specifically. And the sealed estates will be spent once, under the registered protocol with a named external witness, at which point the opinion figures in Section 2.5 acquire a final rather than a developmental status.

The broader claim we would defend is narrow and structural. A model that can lose a pre-registered bet, and does, occupies a different epistemic position from one that cannot be placed in the position of losing. Earth-1's material board lands on real anchors, its census is invariant across a five-hundredfold range of population, its world is recoverable bit-for-bit across independent implementations, and it agrees with real human majorities on five cells in six. It is also, by its own registered gates, behind two statistical baselines on continuous opinion estimation and below its direction target on events. Both halves of that sentence are measurements, taken with the same instrument, under rules written before the answers were known. That is the property the architecture was built to have.

4 Methods

This section describes what exists and how it is evaluated. It is deliberately not a reproduction specification: update equations, fitted constants, functional forms, injector templates, the registered vocabulary and internal module structure are withheld. Where a stated quantity would permit reconstruction of a withheld mapping, it is given at reduced precision or omitted, and the omission is marked. The claims in this paper rest on measured outputs against registered gates and sealed judges rather than on disclosed internals.

4.1 Population substrate

The substrate carries 194 countries. Each country holds a five-dimensional contingency table over sex (2 levels), age band (6), education (3), income (3) and settlement (2), giving 216 cells per country and 41,904 cells in total. Age, sex and settlement margins are census-derived for all 194 countries. Education and income margins are measured directly from survey microdata for 63 countries; for the remaining 131 they are pooled from donors within the same income tier, and the distinction is recorded per country as a provenance flag that travels with any figure computed from that country.

The joint is fitted from its margins by iterative proportional fitting over pooled donors[20]. The alternative of assuming independence between the demographic axes was evaluated as a baseline on withheld joint tables and rejected: IPF beat independent marginals by 24.8% at $p \leq 2.4 \times 10^{-13}$. The practical content of that margin is that the correlation between education, income and settlement is a measured property of each country rather than an artefact of the fitting procedure.

Adult age structure is not drawn parametrically. It is derived from stable-population density using each country's own Gompertz survival curve[21], so that the age pyramid and the mortality process are consistent with one another by construction rather than by calibration.

Aggregation is population-true. Every agent carries a weight equal to the ratio of its country's census share to that country's share of the simulated population, so that a global statistic is an estimate of the world rather than of the resident sample. The world population anchor is 8,140,897,523. This is the mechanism by which a 200,000-agent world reports a global poverty rate at all, and it is why Section 2.4 finds aggregate figures invariant in population size.

4.2 The behavioural grammar

Agents carry eight forces. The forces are latent channels through which circumstance becomes disposition and disposition becomes action; they are updated daily by arithmetic over agent state and are not natural-language quantities. Their definitions, couplings, update equations and constants are withheld.

Three structural properties are load-bearing for the results and are stated without disclosing the mechanism. First, the grammar is universal: the same operators with the same constants run in every country, so cross-country variation in outputs follows from variation in circumstance and composition rather than from per-country tuning. Second, forces are moved each day by two competing operations — a social push along graph edges and a restoring pull toward the state implied by the agent's own material circumstances — and the material target is returned rather than applied so that the two can be composed and inspected separately. Third, dispositional traits are themselves updated on a slower loop from the agent's deviation relative to its neighbourhood, so that personality is an outcome of social position as well as an input to it.

4.3 Deterministic tick and world hashing

The world advances in discrete days. Every stochastic draw on the daily path is taken from an explicitly seeded generator, and the engine contains no dependence on wall-clock time, on iteration order over unordered containers, or on any third-party inference service. A world hash is computed over the complete state, and it is this hash that the determinism checks compare.

Determinism is verified along three independent construction paths — rebirth from the same seed, continuation from a saved checkpoint, and a flag-on run against its own baseline. Six of six checks pass, including branch contrast and historical rebirth, and the question-answering layer reproduces a model probability to seven decimal places on demand.

Two controls establish that this is a meaningful rather than a nominal property. In the failing direction, a perturbation of 10^{-9} applied to a single agent's internal well-being value in a four-million-agent world changes the world hash, and undoing the perturbation restores it exactly; the hash is therefore sensitive to one part in a billion of one person's interior state and cannot be agreeing by coarseness. In the re-implementation direction, a second implementation of the daily path — an internally developed laboratory rewrite, compared against the canonical engine at a registered tolerance of bitwise zero — produced 38 of 38 bitwise-identical checkpoints over a full 365 simulated days at 200,000 agents with equal day-365 world hashes, together with 11 identical checkpoints per seed on shorter runs at two further seeds. No physics was altered to obtain any of that agreement. Both code bases are Earthling Labs' own; a third-party port checked through the public hash endpoint is the outstanding stronger test.

A configuration stamp accompanies every run. The physics configuration is recorded and asserted at load, and a run whose stamp does not match the frozen configuration is refused rather than silently executed. This control exists because an earlier scaling campaign was launched on engine defaults rather than the frozen flag set, and produced a materially different equilibrium before the discrepancy was caught; the stamp converts that failure mode from a matter of operator discipline into a hard gate.

4.4 Branch-versus-null contrast

Claims about the effect of a perturbation are produced as a paired contrast rather than as a description of a single run. A world is advanced to a chosen date, then forked: one branch receives the perturbation and the other does not, and both continue under common random numbers[19] so that the stochastic path is shared and cancels in the difference. Effects are reported as the paired difference across seeds with a seed-level error bar.

The engine carries a registered class noise floor below which a branch difference is not reported as a signal. When a perturbation produces a difference beneath that floor, the system returns an abstention naming the structural absence responsible rather than a number. The floor's value is withheld. This is the mechanism by which the fiscal-shock case in Section 2.8 returned a physics abstention instead of a spurious effect.

4.5 Social graph and influence

The social graph is constructed once at genesis with homophilous structure across several relationship channels, each with its own degree and weight distribution, including a small number of high-degree hubs. It is not a summary statistic: social computation reads a living view rebuilt after each day's deaths, migrations and job changes, so that deceased agents retain their edges as history but carry no weight in current dynamics.

Daily influence is dyadic. Each agent draws a small number of encounters sampled by tie weight from its living neighbourhood, and for each encounter moves a fixed fraction of the way toward that partner's current value, scaled by a susceptibility term depending on the agent's own age, conviction, mental state and hunger. The same encounters accumulate agreement evidence, from which conviction is updated at the end of the day. Tie weights are then updated from the day's interaction, and a limited number of agents rewire toward partners closer to them in force space. Fractions, susceptibility functions and rewiring rates are withheld.

One property of the encounter sampler is recorded as a limitation because it bears on how the paper's error bars should be read: the encounter stream is keyed on simulation day and is independent of the world seed. Seed replicates therefore differ in their population draw and graph — the dominant source of variance — but do not resample the encounter sequence itself, so reported variance under-states the contribution of the influence channel specifically. The magnitude of that under-statement is quantifiable by reseeding the stream and is registered as outstanding work.

4.6 Opinion readout and evaluation protocol

Opinion estimates are produced by a readout from agent state to survey-item response. The readout's construction is withheld. Evaluation is leave-one-country-out: the country under evaluation contributes nothing to the estimate for its own items. Items requiring named-entity knowledge that the substrate does not carry trigger abstention, and abstained cells are excluded from both the model and every baseline, so that abstention cannot flatter the comparison.

Three baselines are scored on the identical cell set. The naive baseline assigns the grand mean across countries. The region-copy baseline assigns a country the attitudes of its region. The MrsP baseline is a multilevel regression with poststratification over census covariates, the standard small-area estimator for exactly this task[22,23,24]. Scores are national mean absolute error in percentage points, and majority agreement is reported alongside distance measured in units of the survey's own sampling error, so that a model answer can be compared against the precision of the survey it is being scored against rather than against an arbitrary tolerance.

Estates are separated by design. The development estates are two World Values Survey Wave 7 sets (held-out and never-labelled extended items) and the Pew Global Attitudes frame as compiled in GlobalOpinionQA; these may be scored repeatedly, and every opinion figure in this paper comes from them. Holdout and prospective estates — including a held-out Pew survey wave and a sealed judge split of the Pew-derived compilation — are sealed, registered by name and location, and reserved for a single scored exposure under witnessed final scoring. No Gallup World Poll benchmark appears in this paper: a Gallup aggregate tier was ruled out of scope for the current version by written ruling, pending data acquisition, and is registered as planned work alongside microdata negotiations with additional survey programmes.

4.7 Frozen-forecast protocol and sealed judges

Historical retrodiction follows a fixed order that admits no move after the outcome is visible.

1. A world is born at date *T* from archive material, under a mechanical assertion that no input, anchor or news item in the consumed set postdates *T*. The assertion is executed, not asserted informally; a violation aborts the run.
2. The question, the horizon, the scoring procedure and the pass tiers are written and committed.
3. The judge is registered by name and location, with its outcome values unfetched.
4. The model output is frozen and committed as a hashed artefact.
5. Only then is the judge fetched, and the result is scored once.

Where a judge is discovered to be biased, the correction is applied forward-only and never retroactively, with the raw table reported beside the corrected one. In one instance the event-count judge was found to over-rank countries with dense media coverage, and normalisation by country total event volume was adopted from the third event onward, leaving earlier scored rows as they were recorded.

Model probabilities are reported to four decimal places. The underlying branch distances that generate them are withheld, since publishing both at full precision would disclose the readout mapping exactly.

4.8 Experience loop and control arms

The learning experiment is a sealed synthetic-recovery design, pre-registered in full before any cycle ran. Twenty twin worlds at 20,000 agents were generated against sealed planted parameters — sixteen well-specified and four deliberately misspecified — and run for twenty-four cycles of thirty simulated days each. The learner observes a summary vector per cycle and updates a particle representation over the eligible parameters; the planted values are not visible to it.

Five control arms were registered. The *frozen* arm never updates. The *naive updater* smooths observations without structure. The *shuffled-resolution placebo* receives observations whose correspondence to outcomes has been destroyed, and is the arm that tests whether apparent learning can arise from fitting noise. The *shock-window* gate scores only the cycles containing sharp forcing. The primary arm, added as a binding condition on the pre-registration, is the *informed non-learner*: identical particle machinery and identical observation access, but no persistent update, since per cycle its weights are computed fresh from that window's observation distance alone with parameters fixed at their prior draws. It may condition on what it has just seen; it may not learn. The gate is the conjunction of every registered comparison, and scoring is paired log-CRPS[26] with Wilcoxon signed-rank significance.

Supporting gates cover calibration and honesty: interval coverage against a registered band, parameter-recovery error, and behaviour under deliberate misspecification, where the requirement is that the loop widen rather than narrow its intervals. Replay from the hash-chained ledger is required to be bit-identical before the cycle closes. One recorded caveat is carried forward: the stage scorer omitted the non-learner arm, so the two gates involving it were computed directly from the arm files by the same paired method, with a scorer patch outstanding.

4.9 Scale measurement

Memory and time were measured at 1M, 4M, 16M, 64M and 100M agents on a single 48-core AMD EPYC 9454P. Reported bytes per agent are steady-state resident state divided by population; peak resident memory is reported separately because construction requires roughly 3.2 times the final state as working memory, and it is that transient, not the steady state, which binds a single machine.

The scale-invariance comparison is run as a control: identical configuration and identical seed lineage at four populations, with the census read at the same simulated day. The four-population control was executed on the engine's superseded default configuration; the record initially attributed its census levels to equilibration and later retracted that attribution, identifying the configuration as the cause, so its levels are comparable to no committed board and only the columns' constancy in population is used. Under the frozen configuration the day-5 census is already near-calibrated and drifts gently onto the anchors, and invariance there is evidenced by the 100M-versus-4M tracking. The calibrated board is measured at 180 days under verified frozen physics, and the two kinds of run are kept apart throughout.

Mortality observation deserves a specific note, because the obvious method is wrong. A deceased agent's slot is reused within the same tick, so a naive comparison of the alive mask before and after a tick sees an occupied slot on both sides and misses the death; measured against a known ground truth this method captured 12% of deaths. All mortality figures in this paper are taken with an observer that detects person-identity turnover as well as the alive flag and reads ages from the pre-tick snapshot, which captures 100% on the same test. Where an earlier campaign reported mortality taken by the naive method, those lines are marked VOID in the record rather than restated.

4.10 Governance

Cycles change one named thing at a time, so that any movement in the gates is attributable to it. Gates are fixed by written ruling before the run. VOID is reserved for defects in the measuring instrument and is never invoked for an unwelcome result; where a finding is voided, the instrument defect that voided it is recorded alongside, and where a voided finding is later re-measured in the opposite direction, both readings stay on the record. Final scoring of the sealed estates is currently witnessed by the founder; a named external witness is being arranged for the single scored exposure, for the reason given in Section 3.2.

Freezes carry their failures. The physics tag on which every result in this paper was measured was released as a freeze-with-named-regression, carrying the failed direction gate and five named regressions rather than a fix that would have reopened the calibration loop. A registered substrate defect discovered after the freeze was recorded and deliberately left unpatched for the same reason and scheduled as a later cycle.

No benchmarking against frontier language models is performed, and the reason is scientific rather than procedural. A language-model baseline is conditional on a checkpoint that will be revised or retired; it is not exactly reproducible; and none of its outputs can be attributed to a mechanism. It therefore cannot enter the evidence chain of Section 3.2 — there is no artefact to freeze and nothing a pre-registration can bind — so it is not an admissible comparator for a pre-registered claim, and a score against one would be unreproducible by construction within months. The project's governing document records this as a standing prohibition; the architecture of the evidence chain is why the rule exists. The comparisons drawn in Sections 2.2 and 2.9 are structural — concerning reproducibility, cost and conditionality — and carry no implication about relative accuracy.

4.11 Interfaces

The engine exposes a versioned interface for questions, structured consequence reports, world state and history, event streams, and population-frame discovery, so that the results described here can be exercised against a running world rather than taken on the report's word. Every answer carries its calibration tier, its provenance, and the configuration stamp of the physics that produced it. Answers the physics cannot support are returned with the register's named missing channel rather than as numbers. The interface also serves the full-world hash of the running snapshot together with its configuration stamp, so the determinism claim of Section 2.2 is exercisable by any key holder: an identical reconstruction must reproduce the served value exactly, and a caller who gets a different hash has falsified it.

4.12 Related work

Classical agent-based modelling established that simple local rules can generate non-obvious population structure, from residential segregation[2] and artificial societies with economies and demographics[3] to threshold models of collective behaviour[4], cascades on networks[5] and bounded-confidence opinion dynamics[6,7,30]. Methodological work in the tradition has concentrated on description standards[16], on the persistent difficulty of replicating published models[15], and on the field's role in economics and policy[9,10,31]. Earth-1 belongs to this lineage in its commitment to legible mechanism, and departs from it in grounding the population in a fitted demographic joint and in treating exact reproducibility as a hard requirement rather than an aspiration.

Language-model agent societies begin with generative agents[11] and scale through successive systems to billion-agent populations with survey-derived profiles[1]. This literature has solved problems Earth-1 does not attempt, above all open-ended behaviour expressed in language: an agent that can articulate a reason, hold a free-form discussion, or respond to a scenario posed in words on the day it is invented. Earth-1's agents carry no language at all, and questions of that kind are outside what it can answer.

The estimation baselines against which the opinion readout is scored come from the survey statistics literature, principally multilevel regression with poststratification[22,23] and its subnational extensions[24], with iterative proportional fitting[20] and calibration estimators[32] underlying the population frame itself. Scoring follows standard proper-scoring practice[26,27], and the pre-registration discipline follows the reform literature in the empirical sciences[17,18,28].

4.13 Data and code availability

The population frame is derived from public census and survey sources: national census margins, World Values Survey microdata[33], Pew Research Center survey frames[34], World Bank poverty and income series[35], and United Nations population estimates[36]. Judge datasets used in the retrodiction battery are public and are registered by name, location and content hash in the frozen forecast artefacts.

The engine, its physics, the calibration ladder and the readout construction are proprietary and are not released. Results are exercisable through the interface described in Section 4.11.

5 Acknowledgements

We thank the maintainers of the public statistical infrastructure this work depends on — national census agencies, the World Values Survey Association, the Pew Research Center, the World Bank and the United Nations Population Division — without whose published margins and microdata a census-grounded population frame would not be constructible.

6 Author contributions

P.N. conceived Earth-1, directed the research programme, fixed every gate and ruling before the run it governs, wrote the sealed priors, imposed the binding condition that the learning experiment include an informed non-learner arm, and approved the freeze and this manuscript. Engine implementation, the population substrate, the calibration ladder, the evaluation harness, the interfaces and the drafting of this paper were carried out by the Earthling Labs engineering effort under P.N.'s direction. All measurements reported here were produced under the frozen physics configuration and recorded in the project's operational ledger.

7 Competing interests

The authors are affiliated with Earthling Labs, which develops Earth-1 commercially. The engine, its physics and its readout construction are proprietary.

References

Guan, H., He, J., Fan, L., Ren, Z., He, S., Yu, X., Chen, Y., Xu, X., Zheng, S., Gao, Y., Chen, E., Liu, T.-Y. & Liu, Z. Modeling Earth-scale human-like societies with one billion agents. *arXiv* 2506.12078v2 (2026).

Schelling, T. C. Dynamic models of segregation. *Journal of Mathematical Sociology* **1**, 143–186 (1971).

Epstein, J. M. & Axtell, R. *Growing Artificial Societies: Social Science from the Bottom Up* (MIT Press, 1996).

Granovetter, M. Threshold models of collective behavior. *American Journal of Sociology* **83**, 1420–1443 (1978).

Watts, D. J. A simple model of global cascades on random networks. *Proceedings of the National Academy of Sciences* **99**, 5766–5771 (2002).

Deffuant, G., Neau, D., Amblard, F. & Weisbuch, G. Mixing beliefs among interacting agents. *Advances in Complex Systems* **3**, 87–98 (2000).

Hegselmann, R. & Krause, U. Opinion dynamics and bounded confidence models, analysis, and simulation. *Journal of Artificial Societies and Social Simulation* **5**(3) (2002).

Ferguson, N. M. *et al.* Impact of non-pharmaceutical interventions (NPIs) to reduce COVID-19 mortality and healthcare demand. *Imperial College COVID-19 Response Team, Report 9* (2020).

Bonabeau, E. Agent-based modeling: methods and techniques for simulating human systems. *Proceedings of the National Academy of Sciences* **99**, 7280–7287 (2002).

Farmer, J. D. & Foley, D. The economy needs agent-based modelling. *Nature* **460**, 685–686 (2009).

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P. & Bernstein, M. S. Generative agents: interactive simulacra of human behavior. *Proceedings of UIST '23* (2023).

Popper, K. R. *The Logic of Scientific Discovery* (Hutchinson, 1959).

Peng, R. D. Reproducible research in computational science. *Science* **334**, 1226–1227 (2011).

Donoho, D. L. An invitation to reproducible computational research. *Biostatistics* **11**, 385–388 (2010).

Edmonds, B. & Hales, D. Replication, replication and replication: some hard lessons from model alignment. *Journal of Artificial Societies and Social Simulation* **6**(4) (2003).

Grimm, V. *et al.* A standard protocol for describing individual-based and agent-based models. *Ecological Modelling* **198**, 115–126 (2006).

Nosek, B. A., Ebersole, C. R., DeHaven, A. C. & Mellor, D. T. The preregistration revolution. *Proceedings of the National Academy of Sciences* **115**, 2600–2606 (2018).

Ioannidis, J. P. A. Why most published research findings are false. *PLoS Medicine* **2**, e124 (2005).

Glasserman, P. & Yao, D. D. Some guidelines and guarantees for common random numbers. *Management Science* **38**, 884–908 (1992).

Deming, W. E. & Stephan, F. F. On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *Annals of Mathematical Statistics* **11**, 427–444 (1940).

Gompertz, B. On the nature of the function expressive of the law of human mortality. *Philosophical Transactions of the Royal Society of London* **115**, 513–583 (1825).

Gelman, A. & Little, T. C. Poststratification into many categories using hierarchical logistic regression. *Survey Methodology* **23**, 127–135 (1997).

Park, D. K., Gelman, A. & Bafumi, J. Bayesian multilevel estimation with poststratification: state-level estimates from national polls. *Political Analysis* **12**, 375–385 (2004).

Leemann, L. & Wasserfallen, F. Extending the use and prediction precision of subnational public opinion estimation. *American Journal of Political Science* **61**, 1003–1022 (2017).

Granovetter, M. S. The strength of weak ties. *American Journal of Sociology* **78**, 1360–1380 (1973).

Gneiting, T. & Raftery, A. E. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* **102**, 359–378 (2007).

Brier, G. W. Verification of forecasts expressed in terms of probability. *Monthly Weather Review* **78**, 1–3 (1950).

Tetlock, P. E. *Expert Political Judgment: How Good Is It? How Can We Know?* (Princeton University Press, 2005).

Sen, A. *Poverty and Famines: An Essay on Entitlement and Deprivation* (Clarendon Press, 1981).

Lorenz, J. Continuous opinion dynamics under bounded confidence: a survey. *International Journal of Modern Physics C* **18**, 1819–1838 (2007).

Tesfatsion, L. Agent-based computational economics: a constructive approach to economic theory. In *Handbook of Computational Economics*, Vol. 2 (Elsevier, 2006).

Deville, J.-C. & Särndal, C.-E. Calibration estimators in survey sampling. *Journal of the American Statistical Association* **87**, 376–382 (1992).

Haerpfer, C., Inglehart, R., Moreno, A., Welzel, C., Kizilova, K., Díez-Medrano, J., Lagos, M., Norris, P., Ponarin, E. & Puranen, B. (eds). *World Values Survey: Round Seven* (JD Systems Institute & WVSA Secretariat).

Pew Research Center. *Global Attitudes Survey* series.

World Bank. *Poverty and Inequality Platform*; population indicator SP.POP.TOTL.

United Nations, Department of Economic and Social Affairs, Population Division. *World Population Prospects* (2024).

Durmus, E. *et al.* Towards measuring the representation of subjective global opinions in language models. *arXiv* 2306.16388 (2023).

Extended Data

The tables below carry the per-estate, per-region and per-arm detail behind the headline figures in Section 2. Values are transcribed from the project's operational ledger. Where the ledger records a value as approximate, or as unavailable, that is reproduced rather than filled.

Extended Data Table 1 | Distance from human majorities, by estate. Frozen physics, 200,000 agents, three seeds, leave-one-country-out, named-entity abstention active with abstained cells excluded from both model and baselines. The pooled row is the ledger's own headline block and is not a recomputation of the estate rows. The third estate is the Pew Global Attitudes frame as compiled in GlobalOpinionQA, recorded in the ledger as *goqa_dev* — an alias of the Pew development frame, with the same items and content hash; it carries no per-cell survey n, so noise-relative statistics do not exist for it.

Estate	Cells	Majority agree	Within noise	Median σ	MAE (pp)	Signed (pp)
All estates pooled	21,776	83.5%	14.2%	3.78	10.65	—
World Values Survey, held-out	5,807	84.6%	13.8	3.80	10.84	−0.04
World Values Survey, extended	6,862	83.7%	14.6	3.75	10.32	−0.00
Pew Global Attitudes (GOQA), development	9,107	82.7%	n/a	n/a	10.77	+0.19

The within-noise fraction is computed over the 12,669 cells where survey sample size is known. Two committed artefacts disagree on the number of abstained cells for this run — the scoring record states 2,198 and its plain-language companion states approximately 500 — and the disagreement is carried on the record rather than reconciled, on the principle that a register which quietly harmonises its own sources has stopped being a register.

Extended Data Table 2 | Median distance in sampling errors, by question family and region. Only family-by-region blocks with at least 30 scored cells are reported; an em dash means the block did not reach that threshold and no value exists in the record. Pooled regional extremes quoted in Section 3.3 come from a separate block of the ledger and are not derivable from this matrix.

Family	C. Asia	E. Asia	E. Eur.	Lat. Am.	MENA	N. Am.	Oceania	S. Asia	SE Asia	SSA	W. Eur.
Democracy/governance	3.49	5.15	4.21	3.00	4.57	—	—	4.29	4.19	3.29	3.55
Other	4.52	4.71	3.00	3.49	4.15	4.37	2.16	4.27	4.71	3.67	3.19
Religion/values	4.73	5.11	4.38	3.29	3.90	—	—	4.20	4.37	4.36	3.11
Security	—	—	—	3.54	4.16	—	—	—	—	—	—
Technology/climate	4.44	3.79	2.24	3.11	2.62	—	—	—	4.99	—	1.98

Extended Data Table 3 | Country-level extremes, median distance in sampling errors, countries with at least 40 scored cells.

Closest ten	**σ**	**Farthest ten**	**σ**
United Kingdom	2.04	Bangladesh	4.79
Cyprus	2.16	Thailand	4.88
Australia	2.24	Tunisia	4.97
New Zealand	2.32	China	5.00
Romania	2.48	Myanmar	5.01
Argentina	2.64	Mongolia	5.27
Ecuador	2.69	Egypt	5.58
Uruguay	2.80	Indonesia	6.05
Russia	2.82	Vietnam	6.27
Slovakia	2.88	Tajikistan	7.85

Extended Data Table 4 | The calibrated board reproduced at twenty times the population. Four million agents, 180 days, verified frozen physics, against the 200,000-agent reference board. Every line falls inside its registered gate. The extreme-poverty line reproduces the board including the board's own known miss against its real anchor of 10.4%.

Quantity	**4M × 180 d**	**200k board**	**Difference**
Median income	9.52	9.605	−0.9%
Poverty, $8.30/day	45.29%	45.88%	−0.59 pp
Poverty, $3.00/day	16.30%	16.80%	−0.50 pp
Crude death rate	0.00738	0.00727	+0.00011
Mean age at death	68.29	68.99	−0.70 yr

Extended Data Table 5 | A hundred-million-agent world tracked against a four-million-agent control through the settling window of the frozen configuration, whose day-5 census is already near-calibrated. Poverty gap is the absolute difference in the $8.30/day headcount between the two runs at the same simulated day.

Simulated day	**5**	**15**	**30**	**60**
Poverty gap (pp)	0.01	0.00	0.02	0.06
Mean age at death	—	—	—	identical (67.96)

Extended Data Table 6 | Experience Loop v0.2, recorded skill by arm. Two distinct quantities appear and are labelled as such: a late-window aggregate, which exists in the ledger as prose, and the final value of a twenty-four-point per-cycle skill curve stored in the machine-written report. They are not the same statistic and must not be compared across rows. Lower is better. Curve values are quoted to four decimal places; the report stores them at full precision.

Arm	Quantity	Well-specified	Misspecified
Experiential	late-window score	0.216	—
Informed non-learner	late-window score	1.04	—
Experiential	final curve value	0.1365	0.1512
Naive-blind	final curve value	0.2328	—
Naive-forced	final curve value	0.2288	—
Frozen	final curve value	1.6175	1.5715
Placebo	final curve value	1.9080	3.4808
Informed non-learner	final curve value	not recorded	not recorded

The report file carries no per-cycle curve for the informed non-learner, because the stage scorer as first run omitted that arm; the two gates involving it were computed directly from the arm files by the same paired method. The registered scorer was extended to include the arm the same day, fifty minutes after this report was emitted, so the stored report predates the fix; fifteen of fifteen locally recomputable consistency checks on the stored report pass, and a full re-emission under the extended scorer awaits the archived arm files on the host that holds them.

Extended Data Table 7 | Pre-committed protest-onset register, control arm. The quiet controls against which the positives were scored. Modelled onset probability for the controls averages 0.031 against 0.417 for the positives, a thirteenfold discrimination margin; controls at approximately zero satisfy the registered placebo criterion.

Country	Class	Series A	Series B
Uruguay	quiet	0.250	0.125
Japan	quiet	0.000	0.125
Singapore	quiet	0.125	0.125
Norway	quiet	0.000	0.000
Denmark	quiet	0.000	0.000
New Zealand	quiet	0.000	0.000
Canada	quiet	0.000	0.000
Australia	quiet	0.000	0.000
Ireland	quiet	0.000	0.000
Finland	quiet	0.000	0.000

Extended Data Table 8 | Determinism and re-implementation equivalence. Registered tolerance for the port-equivalence cases is bitwise zero; both implementations are Earthling Labs' own (a laboratory rewrite against the canonical engine), so these cases establish that the physics as specified pins the bits across code bases, with a third-party port the outstanding stronger test. No physics was changed to obtain any of the recorded agreement.

Check	Configuration	Result
Same-seed rebirth	canonical engine	single world hash `49bb7d1d949f28f3`
Save-and-load continuation	canonical engine	same hash
Flag-on against baseline	canonical engine	same hash
Branch common-random-number contrast	canonical engine	pass
Historical rebirth	canonical engine	pass
Adapter determinism	question layer	reproduces 0.4311624 exactly
Perturbed-field control	live 4M world	10^{-9} on one agent changes the hash; undo restores it
Port equivalence, long run	200k, 365 days	38/38 checkpoints bitwise identical; equal day-365 hash
Port equivalence, short runs	two further seeds, 10 days	11/11 checkpoints bitwise identical per seed

Extended Data Table 9 | Experience Loop v0.2: the registered design, frozen before any cycle ran. Each element is reproduced from the pre-registration. Numeric parameter payloads of the planted truths and the shock scenario are elided, marked …, and no row should be read as a complete parameter listing.

Design element	Registered before the run
Binding condition	The baseline set must include an informed non-learner — the learner's observation access, no update rule. Beating that arm is the demonstration.
Power	16 well-specified and 4 misspecified sealed truths, seeds 9201–9220 with …. The predecessor cycle's win was underpowered at 8 worlds.
Coverage dial	Observation-noise inflation halved, after the predecessor overshot its registered coverage band.
Placebo gate re-specification	Paired experiential-versus-placebo log-CRPS with CI excluding zero. The predecessor's coded form wrongly punished the placebo for beating frozen.
Sharp shocks	The registered memory-probe shocks were smoothly decaying and a smoother tracked them. v0.2 forcing is a branch-engine scenario at days 240 and 480 (…), applied identically to truth and to every simulated world.
Informed non-learner, defined	Identical particle machinery and observation access; no persistent update. Per cycle, weights computed fresh from that window's observation distance only — no multiplicative accumulation, no resampling, parameters fixed at prior draws forever. “It may condition on what it just saw; it may not learn.”
Primary gate	Experiential beats the non-learner on late cycles, paired log-CRPS CI excluding zero. This isolates parameter learning from mere state conditioning.
Run shape	24 cycles of 30 simulated days at 20,000 agents, with receipts, replay, blinding, and the coverage, recovery, honesty and replay gates.
Success condition	Declared only on the conjunction of every registered gate.

Extended Data Table 10 | Governance rules in force for every measurement in this paper. These rules are procedural rather than technical, and they are what the frozen-forecast protocol depends on.

Rule	Content	Cost when honoured
One named change per cycle	A cycle changes one thing, so movement in the gates is attributable to it	Slower iteration; coupled fixes cannot be shipped together
Gates before runs	Pass tiers, metrics, horizons and country sets fixed by written ruling before the run they govern	A gate cannot be relaxed once the result is visible
VOID is instrument-only	A result may be voided only for a defect in the measuring apparatus, never for being unwelcome; the defect is recorded alongside	Unfavourable results stand unless the instrument is shown to be broken
Sealed judges	Registered by name and location with outcome values unfetched until the model output is committed; scored once	A judge is spent permanently on first use
Forward-only judge correction	A judge found biased is corrected from the next event onward, never retroactively; the raw table is reported beside the corrected one	Earlier rows keep a known-imperfect judge
Freeze carries its failures	A physics tag ships with its failed gates and named regressions rather than a fix that would reopen the calibration loop	The published tag carries a failed direction gate and five named regressions
Configuration stamp	Physics configuration recorded and asserted at load; a mismatched run is refused rather than executed	Runs abort rather than producing comparable-looking output
No frontier-model benchmarking	Standing prohibition in the project's governing document	No accuracy comparison against language models may be published

Extended Data Table 11 | Population substrate: composition, provenance and declared boundaries. The joint is fitted per country; provenance travels with every figure computed from a country.

Property	Value	Note
Countries	194	—
Joint dimensions	sex, age band, education, income, settlement	$2 \times 6 \times 3 \times 3 \times 2$
Cells per country	216	41,904 cells in total
Census-derived margins	age, sex, settlement	all 194 countries
Survey-measured education and income	63	from survey microdata
Tier-fallback education and income	131	pooled from income-tier donors; registered limitation
Fitting method	iterative proportional fitting over pooled donors	+24.8% over independent marginals on withheld joints, $p \le 2.4 \times 10^{-13}$
Adult age structure	stable-population density under each country's own survival curve	not a parametric draw
World population anchor	8,140,897,523	public population series
Not represented	subnational geography; household frame; ethnicity; religion	registered boundaries, Section 3.3

Extended Data Table 12 | The shock-free accumulation null for the wealth-concentration result. Computed analytically on the untouched genesis arrays of the identified genesis world: every agent keeps its job and accumulates at a constant rate proportional to its wage, for accumulation-rate multipliers spanning zero to the infinite-rate limit in which the wealth distribution becomes the wage distribution itself. Census-weighted statistics at day 120. The measured concentration exceeds every member of the family; the strongest shock-free competitor (saving proportional to income above subsistence, last row before the measurement) still falls 13% and 20% short. The genesis identification reproduces the recorded day-0 ratio to 0.13% and is inferential rather than documentary.

Accumulation regime	p99 / p50	Top-1% share
Genesis (no accumulation)	11.90	0.116
Rate multiplier 0.05	13.40	0.127
Rate multiplier 0.1	14.81	0.133
Rate multiplier 0.25	16.41	0.136
Rate multiplier 0.5	16.76	0.138
Rate multiplier 1	17.24	0.141
Rate multiplier 5	17.54	0.140
Wage-only limit (rate $\to \infty$)	17.63	0.141
Surplus-proportional variant, limit	26.15	0.162
Measured, day 120	**29.48**	**0.194**

Extended Data Table 13 | Wealth-confound re-scoring of the protest-onset register. Income group is the World Bank classification carried per country. The permutation tests are exact within income strata; because positives and controls co-occur in only one stratum (2 positives against 12 controls), the design cannot yield p below 0.011 regardless of effect size, so non-significance here is a statement about the register's construction as much as about the model. The registered fix is an income-matched control set.

Analysis	Statistic	p	n
Pooled register claim (reproduced)	$\rho = 0.552$	0.0052	24
Confound: outcome vs income group	$\rho = -0.815$	$\sim 10^{-6}$	24
Wealth alone: P(onset) vs income group	$\rho = -0.464$	0.022	24
Partial ρ, income controlled	0.339	0.176 (exact)	24
Within high-income stratum, positives vs controls	0.375 vs 0.031	0.143–0.176 (exact)	2 vs 12
Robustness, FY2011 classification	partial $\rho = 0.339$	0.278 (exact)	24
Within positives: P(onset) vs income	$\rho = 0.056$	0.86	12

Glossary of registered terms

The following terms have specific operational meanings in the project's ledger and are used in that sense throughout this paper.

Term	Meaning as registered
Board	The set of material aggregates — poverty, income, mortality, age structure — measured against real anchors at a stated population and horizon.
Branch / null	Two continuations of one world from the same state, one perturbed and one not, advanced under common random numbers so the difference isolates the perturbation.
Class noise floor	The registered threshold below which a branch difference is not reported as a signal. Its value is withheld.
Configuration stamp	A recorded fingerprint of the physics configuration, asserted at load; a mismatched run is refused.
Estate	A partition of evaluation items. Development estates may be scored repeatedly; holdout and prospective estates are sealed and scored once.
Freeze	A physics configuration fixed by ruling, against which results may be compared. A freeze-with-named-regression ships carrying its failed gates.
Frozen forecast	A model output committed as a hashed artefact before the judge that will score it is fetched.
Judge	An external dataset registered by name and location, with outcome values unfetched, used once to score a frozen forecast.
Physics abstention	A returned non-answer, issued when a branch difference falls below the class noise floor, naming the structural absence responsible.
Population-true	Aggregated with per-agent census weights, so a global figure estimates the world rather than the resident sample.
Register	The enumerated list of standing defects, each with a measured consequence and a closure path.
VOID	A result withdrawn because the measuring instrument was defective, never because the result was unwelcome.

Supplementary note: why census weighting is load-bearing

A simulation of two hundred thousand people cannot report a global poverty rate by counting its own poor. Doing so would report the poverty rate of whatever mixture of countries happened to be instantiated, which is a property of the sampling design rather than of the world. Earth-1 avoids this by giving every agent a weight equal to the ratio of its country's share of the real world population to that country's share of the simulated population, so that a country over-represented in memory is proportionally down-weighted in every statistic computed from it.

Three consequences follow, and each is visible in the results. First, aggregate figures become comparable to published international statistics without a correction step, which is what makes the anchors in Table 1 meaningful rather than coincidental. Second, the aggregate is insensitive to population size, which is the mechanism behind the invariance in Figure 1 and Table 3: adding agents refines the within-country estimate without changing the between-country mixture. Third, and least obviously, it changes what a larger world is for. If aggregate accuracy were a function of population, the case for more agents would be accuracy; because it is not, the case for more agents is resolution — subnational structure, narrow segments, and rare events that a small world produces too few of to measure.

The same construction imposes the boundary recorded in Section 3.3. Weights are national, so the substrate can speak about countries and about the world, and cannot currently speak about regions within a country or about households as units. That is a property of the frame rather than of the physics, and closing it requires subnational margins rather than a change to the engine.